TN SKILL PROJECT

Eight-Phase Project Plan & Implementation Document

Project: Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Detail	Value
Platform	ServiceNow
Security / Automation Area	Access Control / Record Security
Core Workflow	Record request → ACL evaluation → Script checks field value → Access allowed or denied
Team	4 members including Team Leader

Project Overview

This project focuses on implementing a Script-Controlled Access Control List (ACL) in ServiceNow to restrict record access according to the value of a selected field. The solution uses an ACL rule with a script condition so that access is evaluated dynamically rather than being granted uniformly to every user who reaches the record.

The eight-phase structure below follows the organization of the supplied project document while adapting the project scope from Standard Laptop order automation to field-value-based record access control.

Core objective: Ensure that a user can access a protected record only when the configured field value satisfies the security condition defined in the ACL script.

Project Overview

The project addresses a record-security requirement in ServiceNow where access must depend on the value stored in a record field. A standard role or unconditional ACL may be insufficient when the business rule requires record-level access to be evaluated dynamically.

The proposed solution is a Script-Controlled ACL. When a user attempts to access a protected record, ServiceNow evaluates the ACL and executes the configured script. The script checks the target field value and sets the access result accordingly.

End-to-end process

1. A user attempts to access a protected ServiceNow record.
2. The relevant table and field-level ACL is evaluated for the requested operation.
3. The ACL script reads the configured field value from the current record.
4. The script compares the field value with the permitted value or rule.
5. If the condition is satisfied, access is allowed; otherwise, access is denied.
6. The result is verified by testing users/records with matching and non-matching field values.

Project configuration focus

- Identify the target table and field whose value controls access.
- Create or configure the appropriate ACL for the required operation.
- Use an ACL script to evaluate the field value dynamically.
- Test both permitted and restricted record scenarios.
- Verify that the access behavior matches the defined security requirement.

Phase 1: Brainstorming & Ideation

Purpose: Identify the record-security problem and convert it into a focused ACL automation idea.

Problem identified

Users may have access to a table or application but still need to be prevented from opening or working with particular records unless a record field contains an allowed value. A static permission check does not by itself express this field-dependent requirement.

User story

As a ServiceNow administrator, I want record access to be controlled by a field value so that users can access only records that satisfy the defined security condition.

Proposed idea

- Use a ServiceNow ACL on the relevant table/record operation.
- Use an ACL script to inspect the current record's controlling field.
- Allow access when the field value matches the configured condition.
- Deny access when the field value does not meet the condition.
- Verify behavior with positive and negative test records.

Expected value

- More precise record-level access control.
- Consistent enforcement of the business security rule.
- Reduced dependence on manual checking.
- Clear separation between permitted and restricted records.
- Better visibility into how field values affect access decisions.

Ideation output

Selected concept: Implement a Script-Controlled ACL that evaluates a record field and grants or denies access based on the field value.

Phase 2: Requirement Analysis

Purpose: Convert the security idea into functional requirements that can be configured and verified in ServiceNow.

Functional requirements

ID	Requirement	Source-aligned implementation
FR-01	A target ServiceNow table/record must be identified.	Select the table containing the protected records.
FR-02	A controlling field must be identified.	Select the field whose value determines access.
FR-03	An ACL must protect the required operation.	Configure the relevant record/table ACL.
FR-04	The ACL must evaluate the controlling field.	ACL script reads the field from the current record.
FR-05	The permitted field value must be defined.	Script compares the field against the configured condition.
FR-06	Matching records must be accessible.	ACL evaluates to true for the permitted condition.
FR-07	Non-matching records must be restricted.	ACL evaluates to false for the restricted condition.
FR-08	The ACL must be active and testable.	Save/activate the rule and validate with test records/users.
FR-09	The final behavior must be documented.	Capture configuration and test evidence.

Non-functional expectations

- Access decisions should be consistent and repeatable.
- The script should use a clear and maintainable condition.
- The protected operation should be explicitly identified.
- Testing should cover both allowed and denied cases.
- The implementation should avoid unsupported assumptions about unrelated permissions.

Requirement-analysis acceptance points

- Target table and controlling field are identified.
- ACL is configured for the intended operation.
- Script evaluates the field value.
- Allowed value produces access.
- Non-allowed value is restricted.
- Final ACL and test results can be inspected.

Phase 3: Project Design

Purpose: Design the ServiceNow security object, access decision sequence, data flow, and verification path before implementation.

High-level architecture

User → Protected Record → ACL Evaluation → ACL Script → Field Value Check → Allow / Deny → Verification

ACL design

Design element	Configured / planned value
Security object	Access Control List (ACL)
Operation	The required protected operation (for example, read)
Target	Selected ServiceNow table/record
Condition	Field-value-based access condition
Script	ACL script checks the current record field value
Decision	Allow when condition is satisfied; otherwise deny
Testing	Positive and negative access scenarios

Script logic design

- Read the controlling field from the current record.
- Compare the value against the permitted value/condition.
- Return a successful ACL result when the comparison passes.
- Return a failed ACL result when the comparison does not pass.
- Keep the script limited to the access decision required by the project.

Design verification path

- Open the target table and locate records with different field values.
- Use a user who reaches the protected operation.
- Attempt access to a record whose field matches the allowed value.
- Attempt access to a record whose field does not match.
- Record the resulting allow/deny behavior and compare it with the requirement.

Design output

A complete logical security flow is defined from record access request through ACL evaluation, field-value checking, access decision, and verification.

Phase 4: Project Planning

Purpose: Organise implementation activities, responsibilities, sequence, and completion checkpoints for a four-member student team.

Suggested 4-member work allocation

Role	Primary responsibility	Deliverables
Team Leader / Member 1	Overall coordination; ACL design; final validation.	ACL design, coordination checklist, final demo readiness.
Member 2	Target table/field identification; ACL configuration.	Table/field setup and ACL configuration evidence.
Member 3	Script development and access-condition verification.	ACL script and positive/negative test evidence.
Member 4	Testing, screenshots/evidence, documentation and PPT support.	Test cases, screenshots, documentation/PPT inputs.

Implementation schedule

Stage	Planned work	Completion evidence
1. Setup	Open ServiceNow and identify the target table, field and security operation.	Environment and target identified.
2. ACL build	Create/configure the ACL and required script condition.	Saved ACL configuration.
3. Activation	Save/activate the ACL and ensure the rule is available for testing.	Active ACL.
4. Test records	Prepare records with permitted and non-permitted field values.	Positive/negative test data.
5. Access test	Attempt access using the relevant user/context.	Observed access results.
6. Verification	Confirm matching values allow and non-matching values restrict access.	Correct security behavior.
7. Documentation	Capture screenshots, steps, results and lessons learned.	Final report/PPT.

Planning checkpoints

- Do not move to final documentation until the ACL is configured and active.
- Run at least one complete allowed and denied access cycle before the demonstration.
- Keep screenshots of the ACL configuration, script, records and observed results.
- Use consistent terminology: ACL, target table, controlling field, ACL script, allow, deny, protected record.

Phase 5: Project Development

Purpose: Implement the Script-Controlled ACL in ServiceNow and connect the access decision to the selected field value.

Development Activity 1 – Identify the target

- Open ServiceNow.
- Identify the table containing the records to be protected.
- Identify the field whose value determines whether the record can be accessed.
- Confirm the operation to protect, such as record read access.

Development Activity 2 – Create/configure the ACL

- Open the Access Control (ACL) configuration area.
- Create or select the ACL for the target table and protected operation.
- Specify any required role/condition prerequisites used by the project.
- Enable the script portion of the ACL.

Development Activity 3 – Configure the ACL script

- Read the controlling field from the current record.
- Compare the field value with the permitted value or rule.
- Return an allow result when the condition is satisfied.
- Return a deny result when the condition is not satisfied.
- Keep the logic focused on the field-value security requirement.

Development Activity 4 – Save and activate

- Save the ACL.
- Ensure the ACL is active/enabled for testing.
- Confirm the rule targets the intended table and operation.

Development Activity 5 – Prepare verification records

- Prepare at least one record with the permitted field value.
- Prepare at least one record with a non-permitted field value.
- Use the same access context when comparing the two outcomes.

Phase 5: Project Development (continued)

Development output

The implemented solution is an active Script-Controlled ACL associated with the selected ServiceNow table and operation. The ACL script evaluates the controlling field and produces the intended allow/deny result.

Configuration verification

Check	Expected configuration/result
Target table	Correct table containing protected records.
Protected operation	Correct ACL operation selected.
Controlling field	Field identified as the access-control input.
Script enabled	ACL contains the field-value decision logic.
Allowed value	Script produces a successful access decision.
Restricted value	Script produces a failed access decision.
ACL status	Rule is active/enabled for testing.

Implementation notes

- The exact field name and permitted value should match the project configuration used in the ServiceNow instance.
- The script should be tested with actual records from the selected table.
- If other ACLs or roles affect the result, document them rather than assuming the script is the only security control.
- Use screenshots that clearly show the ACL target, operation, script and final test behavior.

Development completion criterion

The team can demonstrate that the same access attempt produces different results when the controlling field value changes, consistent with the configured ACL rule.

Phase 6: Project Testing

Purpose: Confirm that the implemented ACL behaves according to the field-value security requirements.

Test scenario

Attempt to access protected records with both permitted and non-permitted controlling field values, then verify that the ACL script produces the expected access behavior.

Test cases

TC	Test action	Expected result
TC-01	Open the target table and locate the protected records.	Target records are available for testing.
TC-02	Identify the controlling field value on a permitted record.	Permitted field value is visible.
TC-03	Attempt the protected operation on the permitted record.	Access is allowed according to the ACL rule.
TC-04	Identify the controlling field value on a restricted record.	Non-permitted field value is visible in test data.
TC-05	Attempt the protected operation on the restricted record.	Access is restricted according to the ACL rule.
TC-06	Inspect the ACL configuration.	Target table and operation are correct.
TC-07	Inspect the ACL script.	Script evaluates the controlling field.
TC-08	Change/use a test record with a different field value.	Access result changes consistently with the configured condition.
TC-09	Repeat the test with the same user/context.	Results remain consistent.

Testing checklist

- ACL is active.
- Target table and operation are correct.
- Controlling field is correctly identified.
- Allowed field value produces the intended access result.
- Non-allowed field value produces the intended restriction.
- ACL script is visible and understandable.
- Final test evidence can be opened and inspected.

Phase 7: Project Documentation

Purpose: Produce a clear record of the project so another student, evaluator, or team member can understand what was built and how it was verified.

Recommended documentation structure

1. **Project Title** — Script-Controlled ACL – Restrict Record Access Based on Field Value.
2. **Problem Statement** — Explain why record access must depend on a field value.
3. **User Story** — Describe the administrator's need for field-dependent record security.
4. **Project Objectives** — State the access-control and verification objectives.
5. **Requirements** — Document functional requirements and expected outcomes.
6. **System Design** — Show record access → ACL → script → field check → allow/deny.
7. **Implementation** — Document ACL creation, target, operation, script logic and activation.
8. **Testing** — Include test cases, expected results, actual results and screenshots.
9. **Demonstration** — Describe the live sequence used during the demo.
10. **Conclusion** — Summarise how the ACL enforces the field-value access rule.

Evidence / screenshot checklist

- ACL configuration showing the target table and protected operation.
- ACL script showing the field-value access logic.
- Controlling field and permitted value in the test data.
- Test record with the permitted value.
- Test record with the non-permitted value.
- Successful access evidence for the permitted record.
- Restricted/denied access evidence for the non-permitted record.
- Final test results and documentation.

Phase 7: Project Documentation (continued)

Documentation quality rule

Use screenshots that clearly show the relevant field, ACL configuration, script or access result. Keep names and terminology consistent with the configured project. Do not add unsupported technical claims that are not part of the implemented workflow.

Suggested evidence sequence

Evidence	What it should demonstrate
1. Target table	The table containing the protected records.
2. Controlling field	The field whose value determines access.
3. ACL properties	The protected operation and target configuration.
4. ACL script	The field-value comparison logic.
5. Allowed record	A record whose field satisfies the rule.
6. Restricted record	A record whose field does not satisfy the rule.
7. Allowed access result	The expected access outcome for the permitted record.
8. Restricted access result	The expected restriction for the non-permitted record.

Documentation completion criterion

A reviewer should be able to trace the project from the security requirement to the ACL configuration, script logic, field value, and observed access result without relying on undocumented steps.

Phase 8: Project Demonstration

Purpose: Demonstrate the complete security flow and prove that the ACL restricts record access based on the configured field value.

Recommended live demonstration sequence

1. Introduce the problem: some records require access restrictions based on a field value.
2. State the solution: a Script-Controlled ACL evaluates the record field before allowing access.
3. Open the target table and identify the protected records.
4. Show the controlling field and the permitted value.
5. Open the ACL configuration.
6. Show the protected operation and target table.
7. Show the ACL script that evaluates the field value.
8. Attempt access to the permitted record.
9. Show that the permitted record can be accessed according to the rule.
10. Attempt access to the non-permitted record.
11. Show the resulting restriction/denial.
12. Explain that changing the controlling field changes the ACL decision according to the configured condition.

Suggested presentation flow

Slide	Content
1	Project title and team members
2	Problem statement
3	User story and objectives
4	Current problem vs field-based access control
5	System/workflow design
6	ACL configuration and script
7	Target table and controlling field
8	Allowed record test
9	Restricted record test
10	Testing results and screenshots
11	Conclusion and security benefits

Demo success criteria

- The presenter can explain the problem and objective in simple terms.
- The target table and controlling field can be identified.
- The ACL and script can be located and explained.
- A permitted record produces the intended access result.
- A restricted record produces the intended restriction.
- The team can explain how the field value drives the ACL decision.

Phase 8: Project Demonstration (continued)

Final conclusion

The project demonstrates a Script-Controlled ACL in ServiceNow that evaluates a record field value before granting the protected access operation. The ACL script provides a field-dependent security decision, allowing records that satisfy the configured condition and restricting records that do not.

This approach provides a clear mechanism for enforcing record-level access requirements where the field value is part of the security rule. The project also provides a repeatable testing and demonstration path using permitted and non-permitted records.

Team

Member	Role
Member 1	Team Leader
Member 2	Team Member
Member 3	Team Member
Member 4	Team Member

Project completion checklist

- Target table selected.
- Controlling field selected.
- ACL created/configured for the required operation.
- ACL script checks the controlling field.
- Allowed and restricted field values are tested.
- Access results are captured as evidence.
- Four-member team responsibilities are documented.
- Final report/PPT and demonstration are ready.

Source / adaptation note

This document preserves the supplied PDF's eight-phase project-management structure, implementation-document style, testing/documentation flow, and team-allocation section. The project subject has been changed to “Script-Controlled ACL – Restrict Record Access Based on Field Value,” and the team allocation has been changed from three members to four members including the Team Leader. ACL-specific implementation content is newly adapted to the requested project title.